KASONGO TSHIABA Pierre

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Congo, The Democratic Republic/Kinshasa

- ✓ **Digital Health Enthusiast**
- ✓ **Software Engineer**
- ✓ **Data engineer**

I bring over 15 years of experience in designing and building software with over 11 years focused in digital health solutions.
I have special expertise in working with stakeholders to identify, design, develop and deliver digital health solutions using modern technologies across Africa and overseas. I have direct experience with iHRIS, DHIS2, Rapidpro, FHIR and have supported the WHO Workload Indicators of Staffing Needs (WISN) tool and created and implemented a new workforce assessment tool in DRC using the WISN algorithms with interoperable links to iHRIS and DHIS2. As Health Information Systems Developer with IntraHealth, I have led numerous projects, workshops on digital health tools, particularly iHRIS and DHIS2 and their integration in the HIS ecosystem. As active member of iHRIS and DHIS2 communities, I have directly implemented theses tools in DRC, Senegal, Nepal, Ethiopia, Pakistan and have been indirectly supporting other countries and organizations around the globe. I have developed interoperability system to smoothly exchange data between DHIS2 and DATIM/PEPFAR thanks to my experience with DHIS2. As a versatile software engineer, I am highly skilled in software analysis, design, development, and utilization.

SKILLS

- Digital Health
- Open source global goods: DHIS2, iHRIS, openLMIS, PCMT, openMRS, openHIM, openELIS, Kobo, etc.
- Data engineering
- Software engineering: Python, javascript, express, nodejs, reactjs, etc.
- Cloud computing: AWS, GCP
- Database design and management: SQL, Mysql, postgresql, mongodb, elastic search,

WORK HISTORY

September 2025 – now

Independent/ **Consultant**

Keywords: Digital Health, Software engineering, Data engineering.

Provide digital health, data engineering and software engineering consultancy work across the globe.

Mentoring digital health enthusiast

September 2024 – September 2025

BlueSquare/ **Technical Assistant**

Keywords: Digital Health, Software engineering, project management, interoperability, enterprise architecture.

Provide technical assistance to ANICNS for capacity building in Health Information Systems.

Design and implement the interoperability layer for data exchange using FHIR HL7 profiles.

Guide EMR developers to integrate the interoperability layer by implementing FHIR.

Customizing openMRS for HIV patients tracking

Customizing openMRS as National EMR.

Develop interoperable systems and their integration into the enterprise architecture.

Provide assistance in the use of data for decision-making.

December 2023 – Sept 2024

iplus solutions/ Digital Advisor/**PROSANI USAID**

Keywords: Digital Health, project management, supply chain, responsible for the pilot implementation of Medexis within the PROSANI USAID consortium.

Identify the main stakeholders in the digitalization of health in the DRC;

Organize technical meetings to define specifications for identified digital solutions;

Coordinate all activities for the implementation of last mile digitalization following the vision of the project and alignment with the ambitions of the Ministry of Health;

Participate, together with the Consultants and the Ministry of Public Health, in the implementation of the roadmap of the online learning platform on the

management of the supply chain of health products, (AGAPS DRC) , intended for personnel of the Ministry of Health, private sector and other PTFs;

Document and archive the approaches developed in the implementation of innovative strategies that can contribute to the performance of the last mile of the program's drug supply chain

April 2022 – July 2023

Pendulum/ Data Interoperability/ ML Engineer **(fully remote)**

Keywords: Python fastapi, backend, postgresql, javascript, AWS, Ubuntu, DHIS2, OpenLMIS, mSupply.

Design company flagship interoperability layer for DHIS2, mSupply, openLMIS and other LMIS systems to pull data for Machine Learning.

Decide and design generic database schema to standardize our system deployment.

Build the interoperability layer using Python Fastapi, javascript, postgresql and celery for task queuing.

Build, automate and deploy data pipelines that interoperate with country systems to pull data periodically and run Machine Learning algorithm for forecasting.

Conduct complete data analysis to get insight from data pulled from different sources.

Design effective data storage for easy access and privacy.

July 2021 – April 2022

Regional Health Software Engineer/ Digital Health/Intrahealth **(fully remote with trips)**

Keywords: **VueJS, nodejs, postgresql, keycloak, elastic search, FHIR, AWS, Jira, Ubuntu**

Lead and support various digital health(opensource), mainly iHRIS (ihris.org) which is an opensource and customizable Human Resource Information System for Health used across the world.

- Support DHIS2 implementation in Francophone Africa

  Support countries and organisation across Africa implement DHIS2, train users and maintainers.

- IHRIS Core developer

  Ihris version 4 was developped using LAMPP stack, we had to move to MERN stack with iHRIS 5.

Ihris 5 was fully built on top of FHIR as backend. Forms and views where customizable using custom FHIR Profiles. I worked on the backed and frontend to turn this software to FHIR.

- Lead iHRIS implementation Pakistan
  Fund: Gates foundation
  I work with the team of developers from the Ministry of Health as well as the gate foundation team to set up the health human resources management system based on the open source iHRIS system. I customize the system according to the needs of the Ministry, I mentor local developers and advise the team on the design of the system.

- Support iHRIS implementation Ethiopia
  Fund: USAID
  Set up the health human resources system within the Ministry of Health using iHRIS. Customize the system according to country requirements and mentor the local dev team on system architecture and implementation.

- Support DHIS2 implementation for BHBC project in Senegal
  Fund: USAID
  set up the DHIS2 tracker system for monitoring HTA patients. For the production of complex indicators, I also designed a related system with R-script that interoperates with DHIS2 to produce in real time the indicators monitored by the project. I also trained the M&E team in the use of the system.

- Work with other engineers, designers, program staff, and country stakeholders to manage the requirements gathering and documentation and communicate effectively across these diverse teams Work with other engineers, designers, program staff, and country stakeholders to manage the requirements gathering and documentation and communicate effectively across these diverse teams
- Contribute to the design and development of intrahealth flagship softwares
- Support iHRIS software global community

September 2018 – July 2021 **Health Information Systems Developer**/Intrahealth

Keywords: PHP, Nodejs, Vuejs, Mysql, Postgresql, apache, AWS, Jira, Launchpad, Ubuntu

Develop software for the integrated HIV Project (IHAP) f
Fund: USAID

- Implement DHIS2 software as data repository for the entire HIV project's data.

- Implement tracker modules for Mother and children follow-up, self test and ARV distribution.
- Implement tracker for OVC and configure capture app on mobile devices.
- Develop middleware for interoperability between DHIS2 and DATIM/PEPFAR using API.
- Develop a facility registry mapping system for health centers geo-localization.
- Implement an sms system platform for patients appointment reminder
- Data analysis for decision-making
- Administer cloud servers and database.
- Implementation of kobo system for HIV service quality evaluation. This system is used to get feedback from HIV clients on service offered by facilities.
- Implementation of an SMS system for reminding patients appointments

IntraHealth Digital Health

- Develop Workforce Pressure Calculator software to calculate Health workforce need based on workload.

  This is a light version of WISN software by the WHO, this system helps estimate Health worforce need based on workload.

- Work with PATH team to Implement Mhero(SMS system) for COVID Response in DRC. Work with local telcos(vodacom, airtel) to integrate the system with their SMS centers.

December 2019

Health Information System Consultant Advisor consultant: Ministry of Health/Nepal

Implement National Health Worker Registry using iHRIS open source system.

Fund: WHO

- Work with Nepal team to design the system
- Customize Ihris to fit country needs
- Implement a data exchange system (openHim) for data exchange between systems)
- Train developers on system customization,
- Release user manual, SOP and technical documentation

November 2017 – September 2018

iHRIS /Palladium

Expert consultant for Senegal

Implement iHRIS software for HRH2030 project

Fund: USAID

- Audit existing system system to find
- Customize the system according the MoH needs
- Customize dashboard with indicators for decision-making
- Mentor local team on iHRIS customization
- Write documentation and SOPs

November 2014 – September 2018 **Health Information Systems Developer**/Intrahealth

Keywords: PHP, XML, javasript, mysql, launchpad, Ubuntu.

Implement iHRIS software for Primary Health Care Project
Fund: DFID

- Implement iHRIS software (open-source) for digitization of Ministry of Health Human Resources.
- Develop a fingerprint capturing software
- Develop interoperability between iHRIS and DHIS2.
- Develop iHRIS for tracking information on Ebola fever's staffs.
- Train Ministry of Health end-users all over the country.
- Develop software to calculate workforce pressure in health facilities.
- Conduct WISN study to determine staffing needs based on workload.
- Collect, analyze and process data for the WISN study.
- Develop software (WISN Loader) to compile data into WHO WISN software.
- Data analysis for decision-making
- Administer cloud servers, applications and database (AWS).
- Listen and implement users' requirements
- Support other country remotely in implementing iHRIS.
- Participate in staff recruiting process (writing assessments and interviewing)

- January 2011 – November 2014 **Software Developer** /Microcom

Head of Information System Department to computerize the company processes.

Keywords: PHP, jquery, Java, Mysql, apache, Fedora

- Collect users' requirement
- Write specification books
- Develop HRIS with Payroll and attendance modules.
- Develop AAA Billing software.
- Develop CRM information system.

- Deploying AAA billing software in Italy in Eutelsat datacenter.
- Train end-users
- Write user manuals
- Administer databases, applications hardware.
- Recruiting staffs for the department (from test, selection to full integration)

LANGUAGE PROFICIENCY

- French
- English